

Assessment Brief 2025-26

Module Leader: Soumya Sankar Basu		Level: 6
Module Name: Contemporary Software Engineering		Module Code: 55-607819-BF-20256
Assignment Title: Developing a Software Engineering Methodology		
Individual / Group Individual	Weighting: 100%	Magnitude: <i>wordcount/length of...</i> 2500 words or equivalent. This is equivalent of 20 person hours effort. Word count should not be taken literally
Submission date/time: 30 th April 2026/ 3 PM	Blackboard submission: Yes Turnitin submission: No	Format: Word, Excel, Models, Pictures
Planned feedback date: 21 st May 2026	Mode of feedback: Written	
In this assessment are students asked to consider:	Inclusivity and accessibility	Yes
	Sustainability	Not applicable
<u>Module Learning Outcomes</u>		
- LO1 : Able to ideate using standard techniques like Sketching, Brainstorming, Brain writing, Worst Possible Idea etc. - LO2 : Prioritize requirements and identify key requirements for the development of Minimum Viable Product using a design thinking approach. - LO3 : Critically evaluate and adopt contemporary Software Engineering practices and tools to develop this product. - LO4 : Reflect on these practices and how they can be implemented and developed to create a working software engineering process for an inexperienced organisation		
Indicative employability skills you will develop during this module: Problem Solving, Design Thinking, Software Engineering Method, Technical Communication, Team based learning,		

Assessment Brief

This assessment is designed to evaluate your understanding about software engineering method focussing on analysing a problem, arriving at a solution and an initial solution architecture.

You will form your own organisation with 4 members. Tutor will play the role of a client for your organisations. The organisation will be formed during the first teaching week. Your organisation will choose to solve one of the problems given in the supplementary document (CSE Coursework Supplement -2025-26). Problems are picked up from domains what you have experienced personally.

As part of the coursework, your organisation will arrive at an innovative solution to the problem using contemporary techniques like 'design thinking' and decide on set of minimum viable products

You will work with your organisation and individually

- a) arrive at a software engineering methodology that your organisation can follow to develop the solution. The methodology will consist of a development lifecycle, the practices you would follow, the tools you would use to implement the practices, the artifacts that you would produce during development with a probable template. Your choices would need justification.
- b) arrive at an initial solution architecture and technical development environment with justification

Additionally, you need to provide an individual reflection. It needs to reflect on your learning through the assessment. You will need to use your first minimum viable product to validate your software engineering methodology and the initial architecture. You also need to reflect on the practice and tool selection for your software engineering methodology and critically discuss about your selection.

Key Notes:

- In the first week you need to form/join your organisation (4 members) .
- Your attendance is key to succeed in this module. We will aim finish the work within the sessions and will have regular playback to give you opportunity to receive formative feedback.

Deliverable

Your deliverable would have two parts. The first part will be a group one, and the second part will be individual.

Group deliverable will include design thinking deliverables, Minimum Viable Product Definitions and implementation plan for them. You are free to use any tool decided by your organisation to create your deliverable. Paper based deliverables need to be scanned and uploaded. Other deliverables should be submitted in either Word or PDF format, combined in a single ZIP or 7z file.

You will be provided a template for group submission that you may leverage. You are free to use any other template containing the required information. Length of group submission should be equivalent to 700 words.

Individual submission will include software engineering methodology your organisation will follow to develop the solution, the initial solution architecture describing the technical solution to the problem and an individual reflection.

Individual reflection needs to validate the initial architecture using the first minimum viable product and comment on the architecture. It needs to reflect on your practice and tool selection for your software engineering methodology. You will also need to provide your learnings from the module.

You will be provided a template for individual submission that you may leverage. You are free to use any other template containing the required information. Length of individual submission should be equivalent to 1800 words.

You must also upload all deliverables to the module Blackboard site by the deadline.

Writing Style and Word Limit

- The word count excludes images, models or pictures.
- It is important that your submission contain a reference list that contains citations to external sources that may have been used. The citations should conform to the APA referencing system¹.
- The word count of your submission should not exceed the word count as stated above, excluding references and appendices.

Submission

You should submit two different documents in the designated submission point. The first one is your group output (Design Thinking outcome) and the second one is your individual submission including solution architecture, software engineering method and reflection.

¹ [Referencing Guide - Library Online - LibGuides at Sheffield Hallam University](#)

The submission deadline is 30th April 2026 at 4 PM.

Marking

Each member of an organisation will get equal mark by default for the group submission. However, the lack of contribution by individual members (assessed via tutor observation during session through in class engagement & playbacks during the class) may result in some members being awarded proportionately lower marks.

Individual submission would be marked individually.

Artificial Intelligence and Academic Integrity – AI&AI

It is important that if you use AI tools to generate an assignment that you do not submit it as if it were your own work. You must check the AI Transparency Scale (AITS) to determine what level of AI use is allowed.

Our regulations state:

Contract cheating/concerns over authorship: This form of misconduct involves another person (or artificial intelligence) creating the assignment which you then submit as your own. Examples of this sort of misconduct include: buying an assignment from an 'essay mill'/professional writer; submitting an assignment which you have downloaded from a file-sharing site; acquiring an essay from another student or family member and submitting it as your own; attempting to pass off work created by artificial intelligence as your own. These activities show a clear intention to deceive the marker and are treated as misconduct.

Further guidance is available here: <https://blogs.shu.ac.uk/assessment4students/preparing-to-submit-work/#AI>

AI Transparency Scale (AITS)

For this assessment, students cannot use AI as in AITS level 1. All students are required to include a transparency declaration statement which can be added as an appendix to the assessment (not included in the word count)

SHU AI Transparency Scale (AITS)				
AITS	Descriptor	Transparency Statement	AI Contributions	Human Contribution
1	No AI	Artificial Intelligence (AI) has not been used for any part of the activity.	AI is not used for any part of the activity.	All aspects of the activity are human generated, created, edited, and developed.
2	AI for Shaping	AI has been used to shape the initial and/or final parts of the activity.	AI is used for shaping parts of the activity. This includes initial outlining, concept development, prompting thinking, and/or improving structure/quality of the final output.	Most of the activity is human developed/generated. AI ideas and suggestions are refined and reviewed. AI outputs are used for discrete and specific goals/outcomes.
3	AI for Developing	AI has been directed for enhanced development of concepts and outputs.	AI is used to undertake detailed development of many or most aspects of an activity and outputs of that activity.	The human takes a significant role in the enhancement, refinement, and critical review of AI generated elements, combining or curating for any outputs.
4	AI for Enhancing	AI has been implemented for all elements of the task.	AI is used extensively throughout the task to achieve goals and outcomes.	The human directs the use of AI for effective outcomes within an activity. Critical thinking is evidenced for any outputs.

5	AI for Innovating	AI has been used for all elements of a task or piece of work, and it has been used in new, creative, and innovative ways through advanced techniques.	AI is implemented in an advanced and innovative way throughout all aspects of the activity.	AI is used creatively and critically by the human. The human uses AI as a co-creator with a critical thinking approach to generating novel AI activities and outputs.
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Transparency declaration statements

Your statement should:

- Indicate the AITS descriptor you are using.
- Briefly describe how AI was used (if applicable) or confirm no AI was used.

Please include this statement as an appendix in your assessment. Failure to comply with this requirement may be considered a breach of academic integrity under our Academic Conduct Policy.

Assessment Criteria

	Fail (insufficient)			THIRD (sufficient)			LOWER SECOND (good)			UPPER SECOND (very good)			FIRST (excellent)			
Criteria and weighting	Low	Mid	Borderline	Low 3rd	Mid 3rd	High 3rd	Low 2.2	Mid 2.2	High 2.2	Low 2.1	Mid 2.1	High 2.1	Low 1st	Mid 1st	High 1st	Exceptional 1st
C1. Solution Identification using Design Thinking (Group outcome) 30%	Some basic steps of Design thinking have been demonstrated without presenting any meaningful outcome.			A Minimum Viable Product has been identified using Design Thinking techniques.			Need statements have been identified through basic demonstration of envision, ideate, and prioritise. A Minimum Viable Product has been identified.			Need statements have been identified through reasonable demonstration of envision, ideate, and prioritise. Multiple Minimum Viable Products have been identified.			Need statements have been identified through thorough demonstration of six design thinking steps especially empathy map, ideation, and prioritising. MVPs have been identified with proper rationale.			Need statements have been identified through thorough demonstration of six design thinking steps especially empathy map, ideation, and prioritising. MVPs have been identified with proper rationale. A development plan for the minimum viable products has been presented. Appropriate research has been carried out beyond what is taught in the class.
✓																
C2. Solution Architecture 20%	Very basic mention about technology choices (e.g., react for front end, SQLServer for back end etc.).			Technology choices are illustrated. Different layers of the solution are explained diagrammatically (e.g., a basic Architecture Overview).			Technology choices are provided Different layers of the solution are explained diagrammatically with the responsibilities of each layer (a better Architecture Overview).			Technology choices are provided with rationale (why a particular technology has been chosen). Different layers of the solution are explained diagrammatically with the responsibilities of each layer.			Technology choices are provided with rationale. Different layers of the solution are explained diagrammatically with the responsibilities of each layer. Communication between layers is explained.			Technology choices are provided with rationale. Different layers of the solution are explained diagrammatically with the responsibilities of each layer along with technology placement and inter layer communication protocol (extensive Architecture Overview). Relevant reference architecture has been researched and the knowledge acquired it leveraged with proper referencing.

Overall, Marks
100%

University Grade Descriptor (UGD)

Level 6 - Generic grade descriptor: relationship of degree classification to Grade Point and equivalent percentage

Class	Category	Mark range	%	General Characteristics
1 st	Exceptional 1st	93 - 100	96	Exceptional breadth and depth of knowledge and understanding of the area of study, significantly beyond what has been taught in all areas ; evidence of extensive and appropriate selection and critical evaluation/synthesis/analysis and of reading/research beyond the prescribed range, in both breadth and depth, to advance work/direct arguments; excellent communication; performance deemed to be beyond expectation. Work at publishable or commercial standard. The ability to make decisions and systematically carry out tasks/processes with autonomy in unpredictable situations; exercise of initiative in the completion of practical tasks; exceptional leadership skills and evidence of personal responsibility in group contexts; creative flair; extremely well-developed problem-solving skills; the ability to carry out sustained critical reflection on practical work within the wider context of industry/workplace. Exceeds expectations set by the industry/employment context.
1 st	High 1st	85 - 92	89	Excellent knowledge and understanding of the area of study as the student is typically able to go beyond what has been taught (particularly for a mid/high 1st) ; evidence of extensive and appropriate selection and critical evaluation/synthesis/analysis of reading/research beyond the prescribed range, to advance work/direct arguments; excellent communication; performance deemed beyond expectation of the level. The ability to make decisions and carry out tasks/processes with a high level of autonomy; creative flair and the ability to (re)interpret predefined rules/conventions to select and justify individual working practice; excellent problem-solving skills; accuracy and fluency; excellent command of skills appropriate to the task; the ability to reflect critically on practical work within the wider context of industry/workplace. Meets expectations set by the industry/employment context.
	Mid 1st	78 - 84	81	
	Low 1st	70 - 77	74	
2.1	High 2.1	67 - 69	68	Very good knowledge and understanding of the area of study as the student is typically able to relate facts/concepts together with some ability to apply to known/taught contexts ; evidence of appropriate selection and critical evaluation of reading/research, some beyond the prescribed range, may rely on set sources to advance work/direct arguments; demonstrates autonomy in approach to learning; strong communication skills. Broadly autonomous completion of practical tasks/processes; ability to adapt in response to change or unexpected experiences; technical/artistic decision making is very highly developed; a clear command of the skills relevant to the task/process; ability to reflect on practical work and set future goals within the wider context of industry/workplace. Adherence to standards set by the industry/employment context.
	Mid 2.1	64 - 66	65	
	Low 2.1	60 - 63	62	
2.2	High 2.2	57 - 59	58	Good knowledge and understanding of the area of study balanced towards the descriptive rather than critical or analytical ; evidence of appropriate selection and evaluation of reading/research, some may be beyond the prescribed range, but generally reliant on set sources to advance work/direct arguments; communication shows clarity, but structure may not always be coherent. A confident approach to practical tasks; a solid grasp of the related processes, tools, technology; creativity in the completion of the task; proficiency is demonstrated by an accurate and well-coordinated performance; tasks are completed with a good level of independent thought and autonomy; an ability to reflect on practical work and set future goals. General adherence to standards set by the industry/employment context.
	Mid 2.2	54 - 56	55	
	Low 2.2	50 - 53	52	
3 rd	High 3rd	47 - 49	48	Knowledge and understanding sufficient to deal with terminology, basic facts and concepts but fails to make meaningful synthesis; some ability to select and evaluate reading/research however work may be more generally descriptive; general reliance on set sources to advance work; arguments may be weak or poorly constructed; communication/presentation is generally competent but with some weaknesses. Competence in technical/artistic skills; tasks/processes
	Mid 3rd	44 - 46	45	

	Low 3rd	40 - 43	40	are completed with a degree of proficiency and confidence; tasks are completed with a sufficient level of independent thought; effective judgements have been made; evaluation and analysis of performance in practical tasks is evident. Errors in workflow or completion of the task; general adherence to appropriate rules/conventions set by the industry/employment context.
Fail	Borderline Fail	30 - 39	35	Insufficient knowledge and understanding of the subject and its underlying concepts; some ability to evaluate given reading/research however work is more generally descriptive; naively follows or may ignore set material in development of work ; given brief may be only tangentially addressed or may ignore key aspects of the brief; communication shows limited clarity, poor presentation, structure may not be coherent. Practical tasks are attempted; skill displayed in some areas; there are a significant number of errors; a lack of proficiency in most areas; guidance may be needed to reproduce aspects of the task and/or apply learned skills. Tasks may be incomplete; failure to adhere to some of the rules/conventions set by the industry/employment context.
	Mid Fail	20 - 29	25	
	Low Fail	6-19	10	No evidence of knowledge or understanding of the subject; no understanding of taught concepts, with facts being reproduced in a disjointed or decontextualised manner ; ignores set material in development of work; fails to address the requirements of the brief; lacks basic communication skills. A general level of incompetency in practical tasks; an evident lack of practice; set tasks are not completed; few or no skills relating to tasks are evident. No adherence to rules/conventions set by the industry/employment context.
Zero	Zero	0-5	0	Work not submitted, work of no merit, penalty in some misconduct cases.